Methods for Handling Multiple Outcomes in Health Data

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Abstract

- Clinical studies frequently involve multiple outcomes such as Overall Survival, Progression-Free Survival, recurrence, progression and death.
- These outcomes are often related and may occur sequentially over time.
- Analysing each endpoint separately may fail to capture the complete disease pathway.
- This project compares statistical strategies for handling multiple outcomes in health data using the SECOMBIT clinical trial dataset.
- The methods considered include Logistic Regression, Kaplan-Meier analysis, Log-Rank test, Cox Proportional Hazards Model, Composite Endpoint Analysis, Win Ratio, Net Benefit and Multi-State Models.
- The results suggest that traditional methods are useful for preliminary analysis, but advanced methods provide richer clinical interpretation.
- Multi-State Models were particularly useful for modelling transition-specific treatment effects over time.

Keywords: Multiple Outcomes, Survival Analysis, Composite Endpoints, Win Ratio, Net Benefit, Multi-State Models, SECOMBIT, Health Data Science

1. Introduction

1.1 Background

- Clinical research often involves more than one outcome of interest.
- In oncology and other chronic disease settings, patients may experience several clinically important events during follow-up.
- Common outcomes include:
  - Overall Survival
  - Progression-Free Survival
  - Disease Recurrence
  - Metastasis
  - Death
- Traditional statistical methods often analyse each outcome separately.
- Separate analysis may not fully reflect the patient journey.
- Multiple outcomes provide a more complete understanding of treatment effectiveness.
- For example, disease progression may occur before death, and both outcomes together provide a clearer picture of clinical benefit.

1.2 Why Multiple Outcomes Matter

- Patients may move through several disease stages over time.
- Example disease pathway:
  - Stable Disease
  - Progression
  - Recurrence
  - Death
- These outcomes are not independent.
- Progression may increase the risk of death.
- Treatment effects may differ depending on disease stage.
- Death is usually more clinically important than progression.
- Therefore, methods are needed that can capture:
  - Outcome relationships
  - Outcome hierarchy
  - Timing of events
  - Disease progression pathways

1.3 Challenges in Analysing Multiple Outcomes

- Outcomes may occur sequentially.
- Outcomes may be statistically dependent.
- Different outcomes may have different clinical importance.
- Analysing outcomes separately may ignore patient trajectories.
- Multiple testing may increase the risk of false positive findings.
- Treatment effects may change during follow-up.
- Standard Cox models may be limited if the proportional hazards assumption is violated.
- Composite endpoints may combine outcomes that are not equally important.

1.4 Aim of the Project

- To compare different statistical strategies for handling multiple outcomes in health data.

1.5 Objectives

Primary Objectives

- Analyse Overall Survival.
- Analyse Progression-Free Survival.
- Compare separate outcome and composite outcome approaches.
- Evaluate predictive modelling strategies.

Secondary Objectives

- Explore advanced methods such as Win Ratio.
- Explore Multi-State Models.
- Assess usefulness of these methods in healthcare research.
- Compare advantages and disadvantages of each method.
- Identify which methods provide better clinical interpretation.

2. Dataset Description

2.1 SECOMBIT Clinical Trial Dataset

- This project used the SECOMBIT clinical trial dataset.
- SECOMBIT was a phase II randomised clinical trial.
- The study involved patients with metastatic BRAF V600 mutated melanoma.
- The dataset included approximately 209 patients.
- Patients were divided into three treatment arms:
  - Arm A
  - Arm B
  - Arm C

2.2 Main Outcomes

- Overall Survival (OS)
- Progression-Free Survival (PFS)

Overall Survival

- Overall Survival refers to the time from treatment or study entry until death.
- It is one of the most important endpoints in oncology trials.
- It directly reflects patient survival benefit.

Progression-Free Survival

- Progression-Free Survival refers to the time until disease progression or death.
- It captures earlier clinical deterioration before death.
- It is useful when survival follow-up is long or when progression is clinically meaningful.

2.3 Covariates

The main covariates considered were:

- Treatment arm
- Number of metastatic sites
- LDH level
- Tumour Mutational Burden
- JAK mutation status

2.4 Why This Dataset Was Suitable

- It contains multiple clinically important outcomes.
- OS and PFS are related but represent different aspects of treatment benefit.
- It allows traditional survival methods and advanced multi-outcome methods to be compared.
- It includes clinical and biomarker covariates.
- It supports modelling of disease progression from stable disease to progression and death.

3. Literature Review

3.1 Multiple Outcomes in Clinical Research

- Multiple outcomes are common in clinical trials.
- This is especially common in oncology studies.
- Researchers may be interested in whether a treatment:
  - Delays progression
  - Improves survival
  - Reduces recurrence
  - Reduces metastasis
  - Improves clinical decision-making
- These outcomes may not be independent.
- A patient who progresses earlier may have a higher risk of death later.
- Separate analysis of outcomes is simple but may miss relationships between events.
- Advanced statistical methods are needed to evaluate multiple outcomes together.

3.2 Composite Endpoints

What It Is

- Composite endpoints combine several outcomes into one endpoint.
- A patient is considered to have experienced the composite event if any component event occurs.
- For example, progression or death may be combined into a single endpoint.

Advantages

- Increases number of events.
- Can improve statistical power.
- Provides a single summary endpoint.
- Useful when individual event rates are low.
- Can reduce sample size requirements.

Disadvantages

- Components may differ in clinical importance.
- Interpretation can be difficult.
- A treatment effect may be driven by a less important component.
- Important differences between outcomes may be hidden.
- It may not describe the full disease pathway.

3.3 Survival Analysis

What It Is

- Survival analysis is used when the outcome is time to event.
- It is important because not all patients experience the event during follow-up.
- Patients who do not experience the event are censored.

Common Survival Methods

- Kaplan-Meier analysis
- Log-Rank test
- Cox Proportional Hazards Model

Strength

- Survival methods account for timing and censoring.

Limitation

- They usually analyse one outcome at a time.

3.4 Win Ratio

What It Is

- The Win Ratio is a hierarchical method for comparing treatment groups using multiple outcomes.
- The most clinically important outcome is compared first.
- If patients are tied on the first outcome, the next outcome is compared.

Hierarchy Used in This Project

1. Overall Survival
2. Progression-Free Survival

Why It Is Useful

- It respects clinical priority.
- Death is considered more important than progression.
- It avoids treating all endpoints as equally important.

3.5 Net Benefit

What It Is

- Net Benefit measures the overall clinical advantage of a treatment or prediction model.
- A positive Net Benefit suggests that the treatment provides clinical value.
- A negative Net Benefit suggests that the treatment is less favourable.

Why It Is Useful

- It focuses on clinical usefulness rather than only statistical significance.
- It can help compare decision strategies.

3.6 Multi-State Models

What It Is

- Multi-State Models describe patient movement through different health states over time.
- Instead of analysing one endpoint at a time, they model transitions between disease states.

States Used in This Project

- Stable disease
- Progression
- Death

Why It Is Useful

- It captures patient trajectories.
- It models transition-specific treatment effects.
- It accounts for the dependency between outcomes.

4. Methods: How Each Method Works

4.1 Logistic Regression

Purpose

- Logistic regression was used to analyse binary outcomes.
- It evaluates whether an event occurred or did not occur.

How It Works

- The outcome is coded as 0 or 1.
- The model estimates the probability of the event.
- It produces Odds Ratios.
- Odds Ratios compare the odds of an event between groups.

Why Used in This Project

- It was used as a preliminary method.
- It helped assess treatment effects on event occurrence.
- It allowed comparison with time-to-event approaches.

Advantages

- Simple to implement.
- Easy to interpret.
- Allows adjustment for covariates.
- Useful for initial exploratory analysis.

Disadvantages

- Ignores timing of events.
- Does not account for censoring.
- Analyses outcomes individually.
- May be misleading for time-to-event data.

4.2 Kaplan-Meier Analysis

Purpose

- Kaplan-Meier analysis was used to estimate survival probabilities over time.

How It Works

- Uses observed event times.
- Accounts for censored observations.
- Produces survival curves.
- Allows visual comparison between treatment arms.

Why Used in This Project

- It was used to compare OS and PFS between treatment arms.
- It showed survival patterns over follow-up time.

Advantages

- Handles censored data.
- Provides clear visual comparison.
- Non-parametric method.
- Commonly used in clinical research.

Disadvantages

- Does not adjust for covariates.
- Usually compares one outcome at a time.
- Cannot model transition pathways.
- Does not explain how covariates influence risk.

4.3 Log-Rank Test

Purpose

- The Log-Rank test was used to compare survival curves between treatment arms.

How It Works

- It compares observed and expected events between groups.
- It tests whether there is a statistically significant difference between survival curves.
- It produces a chi-square test statistic and p-value.

Why Used in This Project

- It was used to test whether OS and PFS differed between treatment arms.

Advantages

- Simple and widely accepted.
- Useful for comparing Kaplan-Meier curves.
- Easy to report in clinical research.

Disadvantages

- Does not adjust for covariates.
- Less suitable when hazards are not proportional.
- Does not explain treatment effects over time.
- Analyses one endpoint at a time.

4.4 Cox Proportional Hazards Model

Purpose

- The Cox model was used to evaluate the effect of treatment arm and covariates on survival outcomes.

How It Works

- It models the hazard of an event over time.
- It produces Hazard Ratios.
- Hazard Ratios describe relative event risk between groups.
- It incorporates censored observations.
- It can adjust for multiple covariates.

Why Used in This Project

- It was used to assess treatment and biomarker effects on OS and PFS.
- It provided a time-to-event regression framework.

Advantages

- Handles censored time-to-event data.
- Allows covariate adjustment.
- Provides clinically interpretable Hazard Ratios.
- Widely used in clinical trials.

Disadvantages

- Assumes proportional hazards.
- Requires separate models for OS and PFS.
- Does not directly model dependencies between outcomes.
- May be inappropriate if treatment effects change over time.

4.5 Composite Endpoint Analysis

Purpose

- Composite endpoint analysis was used to combine multiple outcomes into a single endpoint.

How It Works

- A composite event occurs if any component outcome occurs.
- For example, progression or death can be combined.
- The combined endpoint is analysed as one outcome.

Why Used in This Project

- It was used to compare separate and combined outcome approaches.
- It helped assess whether combining OS and PFS changed interpretation.

Advantages

- Simplifies analysis.
- Increases number of events.
- Can improve statistical power.
- Useful when multiple outcomes are clinically relevant.

Disadvantages

- Can hide differences between individual outcomes.
- May combine outcomes of unequal clinical importance.
- Interpretation may be less specific.
- The treatment effect may be driven by one component only.

4.6 Win Ratio

Purpose

- The Win Ratio was used to compare treatment arms using a hierarchy of clinical outcomes.

How It Works

- Patients are compared pairwise between treatment groups.
- The most important outcome is compared first.
- If the first outcome is tied, the second outcome is compared.
- Treatment wins and control wins are counted.
- The Win Ratio is calculated as:

$$\text{Win Ratio} = \frac{\text{Number of Treatment Wins}}{\text{Number of Control Wins}}$$

Outcome Hierarchy Used

1. Overall Survival
2. Progression-Free Survival

Why Used in This Project

- OS and PFS have different clinical importance.
- Death is more clinically important than progression.
- The Win Ratio allowed outcomes to be prioritised.

Advantages

- Prioritises clinically important outcomes.
- Incorporates multiple outcomes.
- Avoids treating all outcomes equally.
- Provides clinically meaningful interpretation.

Disadvantages

- More computationally complex.
- Less familiar to some clinical audiences.
- Interpretation requires careful explanation.
- Results may depend on the selected hierarchy.

4.7 Net Benefit

Purpose

- Net Benefit was used to evaluate the overall clinical advantage between treatment groups.

How It Works

- It combines information about benefit and harm.
- Positive values indicate clinical benefit.
- Negative values indicate less benefit.
- Values close to zero suggest limited difference.

Why Used in This Project

- It helped assess whether treatment differences were clinically meaningful.
- It supported interpretation beyond statistical significance.

Advantages

- Focuses on clinical usefulness.
- Supports decision-making.
- Helpful when comparing strategies.
- Easy to interpret when explained clearly.

Disadvantages

- Depends on clinical assumptions.
- Less familiar than standard survival methods.
- Does not replace complete statistical modelling.
- Interpretation can vary depending on the chosen context.

4.8 Multi-State Model

Purpose

- The Multi-State Model was used to model transitions between disease states.

State Structure

- State 1: Stable disease
- State 2: Progression
- State 3: Death

Transitions

- Stable disease to progression
- Progression to death

How It Works

- It estimates the probability of moving from one state to another.
- It estimates transition-specific hazards.
- It evaluates how covariates affect each transition separately.
- It models patient trajectories over time.

Why Used in This Project

- OS and PFS are connected outcomes.
- Disease progression occurs before death for many patients.
- A Multi-State Model can capture the full disease process more realistically.

Advantages

- Captures disease progression.
- Models patient trajectories.
- Estimates transition-specific treatment effects.
- Handles dependency between outcomes.
- Allows dynamic interpretation over time.

Disadvantages

- More complex than standard survival models.
- Requires careful definition of states.
- Requires sufficient event numbers.
- Interpretation can be challenging.
- Results may depend on model assumptions.

5. Methodology

5.1 Data Preparation

- Import SECOMBIT dataset into R.
- Check variable names and coding.
- Identify missing values.
- Recode treatment arm.
- Recode biomarker variables.
- Define OS and PFS event indicators.
- Define composite outcome.
- Prepare time-to-event variables.
- Prepare covariates for regression and survival modelling.

5.2 Descriptive Analysis

- Generate baseline summaries.
- Summarise patient characteristics.
- Compare covariates across treatment arms.
- Check distribution of outcomes.
- Identify number of OS and PFS events.

5.3 Separate Outcome Analysis

- Analyse OS separately.
- Analyse PFS separately.
- Use logistic regression for event occurrence.
- Use Kaplan-Meier curves for survival probabilities.
- Use Log-Rank tests to compare survival curves.
- Use Cox models to estimate treatment effects over time.

5.4 Composite Outcome Analysis

- Create a combined endpoint.
- Define composite event based on progression or death.
- Compare composite outcome results with separate OS and PFS results.
- Assess whether the composite endpoint changes interpretation.

5.5 Advanced Multiple Outcome Analysis

- Apply Win Ratio using OS first and PFS second.
- Apply Net Benefit to estimate clinical advantage.
- Fit Multi-State Model using stable disease, progression and death.
- Estimate transition-specific effects.
- Estimate transition probabilities across treatment arms.

5.6 Model Assessment

- Assess Cox proportional hazards assumption.
- Compare model assumptions across methods.
- Compare interpretability of each method.
- Identify whether traditional methods are sufficient.
- Assess whether advanced methods provide additional clinical insight.

5.7 Comparison Framework

Methods were compared using:

- Ability to handle multiple outcomes.
- Ability to incorporate time.
- Ability to account for censoring.
- Ability to adjust for covariates.
- Clinical interpretability.
- Statistical assumptions.
- Computational complexity.
- Ability to model disease progression.
- Practical usefulness in healthcare research.

9. Project Links

The code repository for this project is available at:

https://github.com/darshu-d/MSc-Research-project-

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